

Enhanced Deepfake and Synthetic Media Detection for Robust Liveness Verification in
Financial KYC using Multi-Modal and Micro-Expression Analysis

DISSERTATION

Submitted in partial fulfillment of the requirements of the
Degree: MTech in Artificial Intelligence & Machine Learning

By

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Dissertation Title : Enhanced Deepfake and Synthetic Media Detection for Robust Liveness Verification in Financial KYC using Multi-Modal and Micro-Micro-Expression Analysis

Name of Supervisor : Omer Shakir Hashmi

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Courses Relevant for the Project & Corresponding Semester:

Sr No.	Course Name	Semester
1.	Computer vision	First Semester (2024-25)
2.	Natural Language Processing Applications	First Semester (2024-25)
3.	Natural Language Processing	Second Semester (2023-24)
4.	Machine Learning	First semester (2023-24)

Abstract

With the increasing sophistication of deepfake technologies and generative AI, identity spoofing through synthetic media has emerged as a significant challenge in financial Know Your Customer (KYC) systems. Traditional liveness detection methods often rely on simplistic cues, such as eye blinking or head movement, which can be easily spoofed by advanced video synthesis models. This dissertation proposes a robust, AI-driven framework that combines **multi-modal analysis** and **micro-expression recognition** to enhance the reliability of liveness verification in financial KYC workflows.

The proposed system will utilize multi-modal inputs—including face video, voice/audio, and possibly depth or infrared data—fused through a deep learning architecture to improve detection accuracy. In particular, **micro-expressions**, which are rapid and involuntary facial movements, will be analysed using convolutional neural networks (CNNs) and attention-based temporal models to detect inconsistencies typical of deepfakes.

These expressions are nearly impossible to replicate authentically in synthetic media, making them highly valuable for detecting spoofing.

A hybrid model combining a micro-expression classifier (e.g., using the SMIC or CASME datasets) and a deepfake detector (e.g., trained on FaceForensics++ or Celeb-DF) will be implemented and evaluated. The system will be tested on benchmark datasets as well as simulated KYC environments. Performance will be measured using metrics like accuracy, precision-recall, AUC, and Equal Error Rate (EER). The project aims to demonstrate how combining modalities and subtle biometric cues can significantly strengthen fraud prevention capabilities in real-world KYC systems.

Key Words:

Deepfake Detection, Liveness Verification, Financial KYC, Multi-Modal Learning, Micro-Expression Analysis, Biometric Security, Synthetic Media, AI Ethics.

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Dissertation Outline (Abstract)

BITS ID No. 2023AB05149

Name of Student: Himanshu Garg

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Designation of Supervisor: Vice President, Engineering Lead

Qualification and Experience: Bachelor of Technology & 15years

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Topic of Dissertation: Enhanced Deepfake and Synthetic Media Detection for Robust Liveness Verification in Financial KYC using Multi-Modal and Micro-Expression Analysis

Himanshu Garg
(Signature of Student)

Date: 21-05-2025

Omer Shakir Hashmi
(Signature of Supervisor)

Date: 21-05-2025

Project Work Title: Enhanced Deepfake and Synthetic Media Detection for Robust Liveness Verification in Financial KYC using Multi-Modal and Micro-Expression Analysis

Discussion on the chosen topic

1. The purpose of the work and expected outcome of the work

The primary purpose of this project is to enhance the robustness of biometric liveness verification systems used in **financial KYC (Know Your Customer)** workflows by integrating **multi-modal deepfake detection** and **micro-expression analysis**. With the increasing threat of synthetic media and deepfakes, especially in identity verification, current KYC systems are vulnerable to manipulation. This dissertation aims to develop an AI-based solution capable of detecting such fraud using subtle physiological and behavioral cues across multiple data modalities.

Expected Outcome:

- A working **multi-modal deepfake detection system**.
- A trained model for **micro-expression analysis**.
- A **fused model** for robust liveness detection tailored for KYC.
- Demonstrated improvements in accuracy and security over unimodal or traditional systems.

2. Literature review done in connection with the work, if applicable

A comprehensive review of the following areas has been conducted:

- **Deepfake detection models** using datasets like FaceForensics++ and Celeb-DF.
- **Micro-expression recognition** techniques using CNNs, 3D-CNNs, and temporal modelling on datasets like SMIC and CASME II.
- **Multi-modal fusion strategies** (early, late, attention-based) for biometric verification.
- Existing research in **face liveness detection**, including anomaly detection, motion analysis, and adversarial training.
- Application of AI in **financial KYC and onboarding** processes for fraud prevention.

3. Brief discussion on the existing process and its limitations

Current financial KYC verification often relies on:

- Passive liveness detection (e.g., eye blink, head turn)
- Manual review of documents or selfie-video submissions

Limitations:

- Vulnerable to spoofing via deepfakes or synthetic videos.
- Lack of behavioural cues like micro-expressions that are hard to fake.
- Poor generalization to adversarial attacks or unseen spoofing techniques.
- Limited use of multi-modal inputs, resulting in single-point failures.

4. Justification for selecting a particular methodology for completing the tasks

- Deep Learning is essential for capturing non-linear relationships and subtle manipulations in video and image data.
- Multi-modal learning enhances robustness by combining different data types (visual, audio, temporal features).
- Micro-expression analysis adds a high-fidelity layer of liveness confirmation difficult to spoof, making it ideal for secure environments.
- Pretrained models (like Xception or EfficientNet) enable efficient transfer learning, saving time and resources.

5. Brief discussion on the Project Work methodology

- Data Collection & Preprocessing: Acquire and align data from benchmark datasets (FaceForensics++, CASME II, SMIC).
- Model Training: Train individual models for deepfake detection and micro-expression recognition.
- Fusion: Combine modalities through neural fusion techniques (e.g., attention-based networks).
- Testing: Evaluate using simulated KYC scenarios with spoofed and genuine samples.
- Metrics: Use accuracy, AUC, EER, precision, and recall to assess performance.

6. Benefits derivable from the work

- Increased security and trust in AI-based digital KYC systems.
- Enhanced liveness detection accuracy using multi-modal evidence
- Contribution to research on deepfake counter measures in finance.
- Potential for real-time deployment in customer onboarding workflows.
- Improved fraud prevention with minimal increase in user friction.

7. Any other details in support of the work

- The proposed system leverages publicly available datasets and tools, making it reproducible and scalable.
- The work aligns with the growing industry and academic focus on ethical AI, biometric security, and anti-spoofing technologies.
- This topic also supports cross-disciplinary learning, combining AI, security, human behavior analysis, and financial technology.

Detailed plan of work

1. Broad Area of Work

Artificial Intelligence, Computer Vision, Biometric Security, Financial Technology (FinTech)

2. Objectives

The primary objectives of this project are:

- To develop a deep learning model capable of detecting deepfake and synthetic media used in identity spoofing.
- To integrate micro-expression analysis as a liveness detection feature.
- To design a multi-modal AI architecture for reliable liveness verification.
- To test and validate the system in a financial KYC scenario.

3. Scope of Work

- Design a multi-modal deepfake and synthetic media detection framework.
- Integrate micro-expression analysis for robust liveness verification.
- Apply the system specifically within the context of financial KYC (Know Your Customer) workflows.
- Prepare and preprocess datasets relevant to deepfakes, micro-expressions, and biometric liveness.
- Develop/ Train machine learning/deep learning models for individual modalities (e.g., face, audio, expressions).
- Implement modality fusion techniques to combine visual, audio, and micro-expression data.
- Train and validate the integrated system using benchmark datasets.
- Evaluate performance using metrics such as accuracy, AUC, precision-recall, and EER.
- Simulate real-world KYC verification scenarios to test system robustness against spoofing.

4. Detailed Plan of Work (for 16 weeks)

Serial Number of Task/Phases	Tasks or subtasks to be done	Planned duration in weeks		Specific Deliverable in terms of the project
1.	Literature review: deepfake detection, liveness verification, micro-expressions, and multi-modal fusion. Identify gaps and finalize datasets.	Weeks 1–2 (2 weeks)		Literature review summary; finalized project scope and dataset list
2.	Collect and preprocess datasets: FaceForensics++, Celeb-DF, SMIC, CASME II, KYC audio/video. Perform face detection, alignment, and normalization.	Weeks 3–4 (2 weeks)		Cleaned and annotated datasets; preprocessing scripts
3.	Develop and validate baseline models: CNN for deepfake detection, 3D-CNN/LSTM for micro-expressions.	Weeks 5–6 (2 weeks)		Working baseline models with validation metrics
4.	Design and implement the multi-modal fusion architecture (early/late/attention-based).	Weeks 7–8 (2 weeks)		Integrated fusion model
5.	Train the fusion model on combined dataset; include tuning and robustness testing.	Weeks 9–10 (2 weeks)		Trained, tuned multi-modal model with training logs
6.	Evaluate the full system in simulated KYC scenarios; test against spoofing attempts. Compute accuracy, AUC, precision, recall, EER.	Weeks 11–12 (2 weeks)		Evaluation report with metrics, graphs, confusion matrix

7.	Draft the full dissertation: Introduction, methodology, experiments, results, discussions. Get supervisor feedback.	Weeks 13–14 (2 weeks)		Draft dissertation submitted for review
8	Final proofreading, formatting, viva prep, and final submission.	Weeks 15–16 (2 weeks)		Final dissertation + viva presentation deck

5. Literature References

The following are referred journals from the preliminary literature review

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Supervisor's Rating of the Technical Quality of this Dissertation Outline

EXCELLENT / GOOD / FAIR/ POOR (Please specify): Excellent

Supervisor's suggestions and remarks about the outline (if applicable).

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